Rotations and complex numbers

Advanced high school · geometry · Progressive exercises · 09

Exercise 1. Find $i(2-3i)$, then its modulus.

Exercise 2. Find the successive images of $1+i$ under four multiplications by i.

Exercise 3. What is the centre of $z' = iz + 2$?

Exercise 4. For $z' = a + i(z-a)$, prove $|z'-a| = |z-a|$.